

When is expensive perception worth paying for?

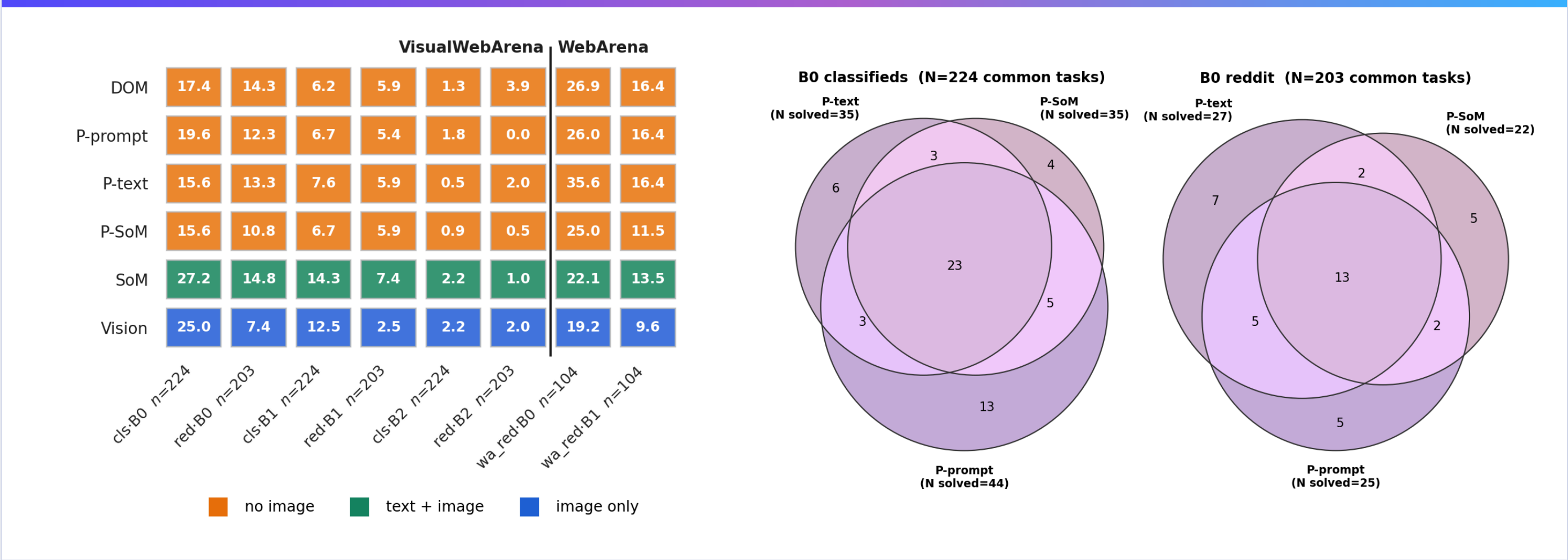
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ONE RECORDED RUN PER VIEW · NOTHING ON THIS SCREEN IS LIVE

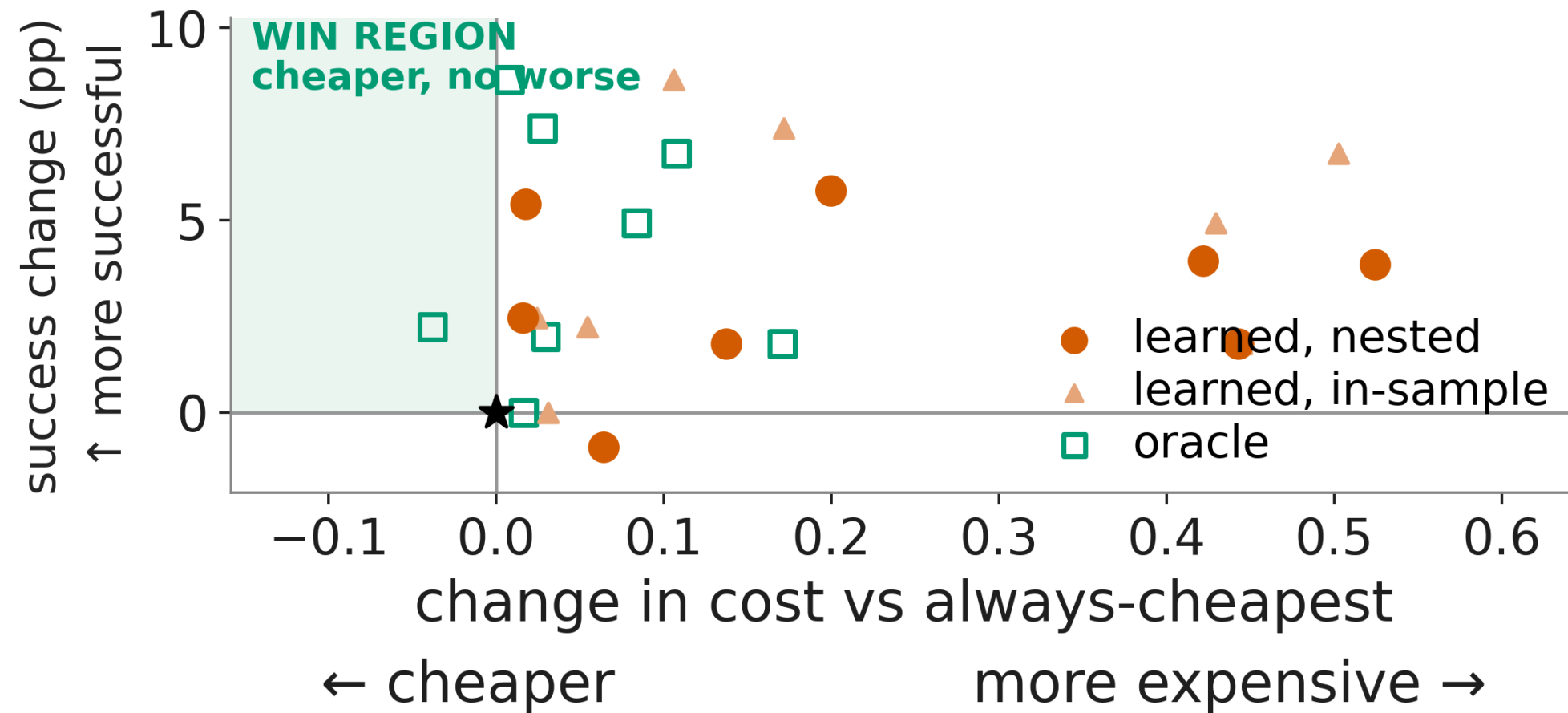
■ Same task, three ways of seeing it, three different bills.

■ Hindsight says choosing pays; a rerun says how much of that is noise.



Hindsight: **+3.45 to +16.35** in 100 vs the best single view · a rerun flips **10–14%**, buys **2.0–7.6**

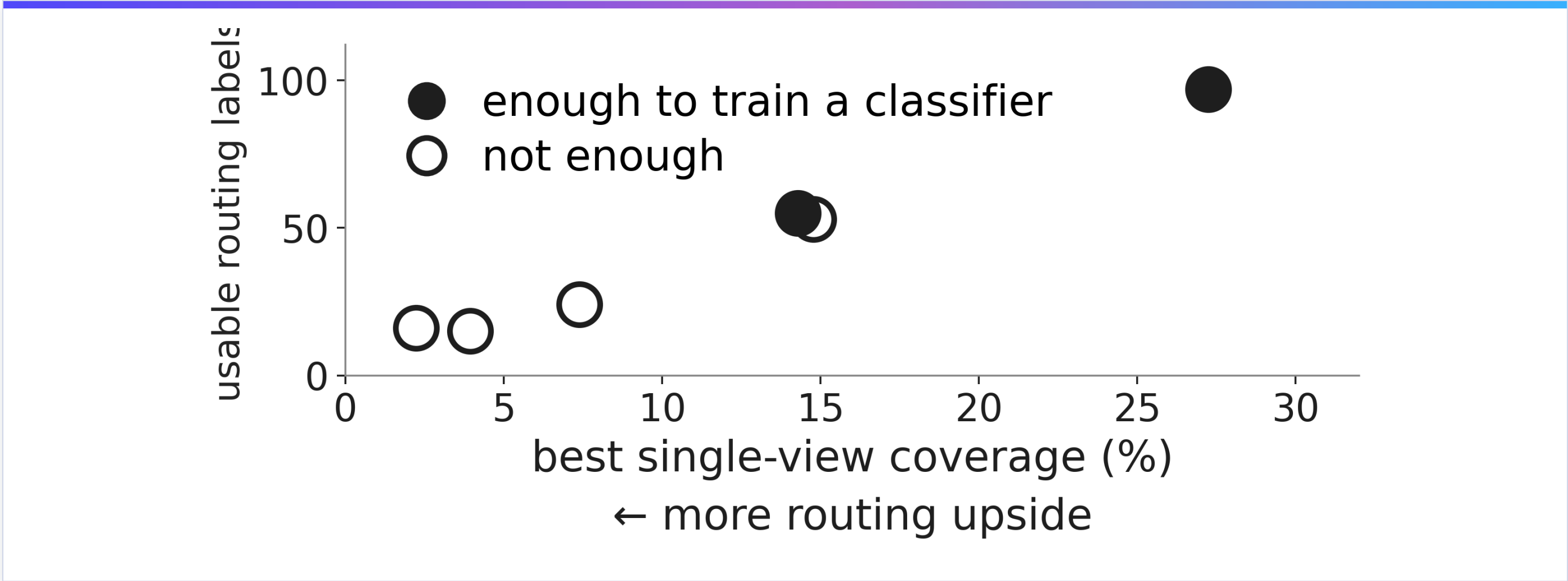
■ Learned routers buy success only by spending more.



Win region = cheaper and no worse. Learned: **0 of 8**. Perfect hindsight: **1 of 8**.

A “WHICH VIEW” EXAMPLE EXISTS ONLY WHEN THE TASK IS SOLVED

■ More routing upside, less usable training signal.



Base success **2–36%** · **15–97** usable examples per setting · enough to train in **2 of the 6** · failing settings need at least **2.1–4.2×** more tasks

■ Improve the agent first, collect reliable examples, then learn when to look.

NOT CLAIMED

- ✗ routing is unlearnable
- ✗ always-cheapest is a floor
- ✗ live runs are scored

THREE ASKS

- ✓ type a task at the board
- ✓ which view do you use, and why?
- ✓ lend us a stronger agent

Look, read, or both?
Hindsight says: choose.
Nothing trained knows in advance.



Come to the board — try your own task, live in all three views.

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BACKUP – TASK 130, RECORDED STEPPING, 39 S